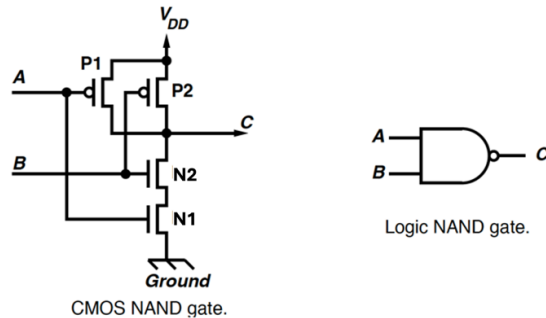


Homework # 3

1. For a two-input CMOS NAND circuit:

- a. Find a two-pattern test for each single-transistor stuck-open fault



Test No.	Fault	Test: Vector 1, Vector 2
1	P1 sop	11, 01
2	P2 sop	11, 10
3	N1 sop	01, 11 or 10, 11 or 00, 11
4	N2 sop	01, 11 or 10, 11 or 00, 11

Notice that the sop faults of N1 and N2 have exactly the same tests. These two faults are equivalent.

- b. Rearrange the eight vectors in a compact set so the sequence is as short as possible

The following sequence of four vectors contains one vector pair for each fault in the above table:

11, 01, 11, 10

Notice that this sequence also detects all single stuck-at faults in the logic model of the NAND gate.

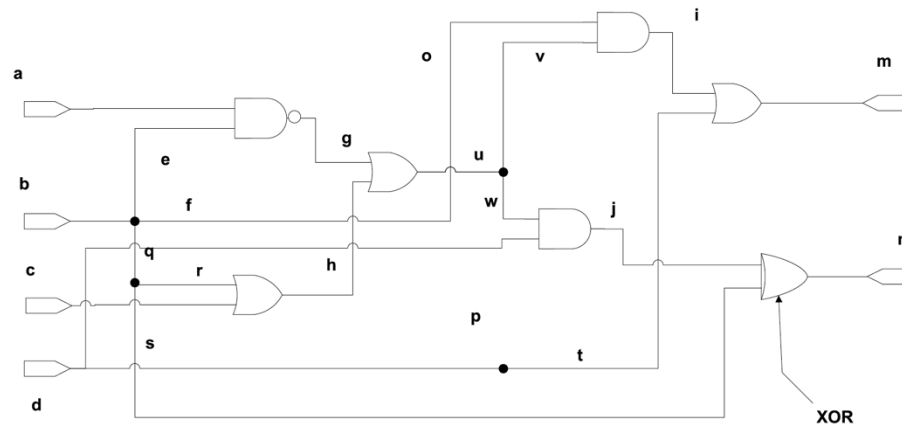
- c. For each stuck-at fault of the NAND gate, find an equivalent transistor (stuck-open (sop), stuck-short (ssh), or a combination) fault.

A stuck-at fault in a signal affects two transistors in the two-input NAND gate. For example, the fault A s-a-1 will mean that N1 remains permanently shorted (N1-ssh) and P1 remains permanently open (P1-sop). The following table gives all equivalences:

Stuck-at fault	Equivalent transistor faults
A s-a-1	N1-ssh and P1-sop
B s-a-1	N1-ssh and P2-sop
C s-a-1	(P1-ssh or P2-ssh) and (N1-sop or N2-sop)
A s-a-0	N1-sop and P1-ssh
B s-a-0	N2-sop and P2-ssh
C s-a-0	N1-ssh, N2-ssh, P1-sop and P2-sop

Notice that the three equivalent faults, A s-a-0, B s-a-0 and C s-a-0, are actually caused by different faulty transistors. They are detected by the same test (11).

2. Show that the two faults a SA0 and c SA1 are equivalent in the circuit below.



Faulty functions for the circuit of Figure 4.6 corresponding to the two faults after logical simplification are:

$$m(a \text{ SA0}) = b + d$$

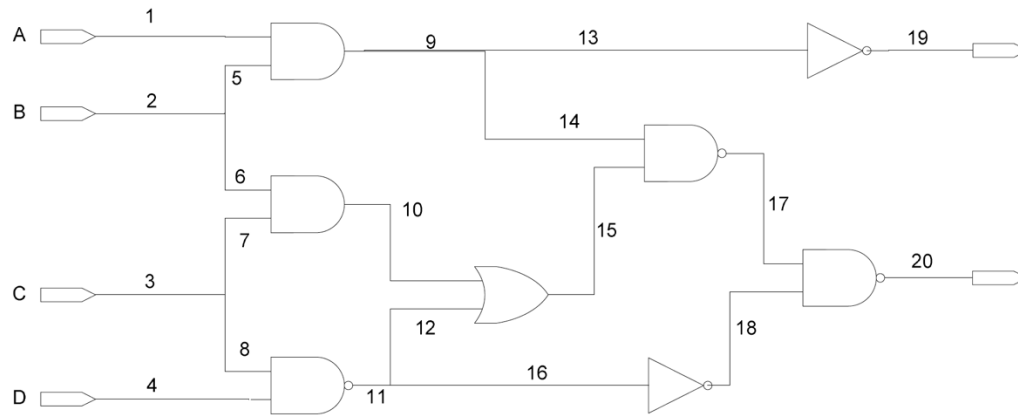
$$n(a \text{ SA0}) = b \oplus d$$

$$m(c \text{ SA1}) = b + d$$

$$n(c \text{ SA1}) = b \oplus d$$

The two faulty functions are indistinguishable and hence the two faults are equivalent. Alternatively, you need to show that the signal $u = 1$ for either of these two faults and these faults reconverge only at line u .

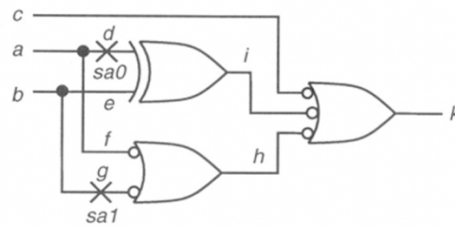
3. For this problem, use the table in the figure in the next page.
- Table 1 gives a fault list for the circuit, after equivalence fault collapsing. Fill in the blanks with equivalent fault sets for each listed fault. (Validity check: the total number of faults in table should be 40)



Fault	List all equivalent Faults
1/1	$\emptyset$
9/0	1/0, 5/0
9/1	$\emptyset$
5/1	$\emptyset$
2/0	$\emptyset$
2/1	$\emptyset$
10/0	6/0, 7/0
6/1	$\emptyset$
7/1	$\emptyset$
3/0	$\emptyset$
3/1	$\emptyset$
11/0	$\emptyset$
11/1	4/0, 8/0
8/1	$\emptyset$
4/1	$\emptyset$
17/1	14/0, 15/0
14/1	$\emptyset$
15/1	10/1, 12/1
12/0	$\emptyset$
18/1	16/0
19/0	13/1
19/1	13/0
20/0	$\emptyset$
20/1	16/1, 17/0, 18/0

Table 1

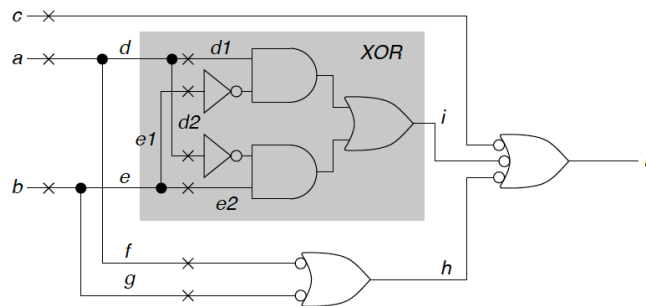
4. For the circuit below:



- Enumerate the minimal set of single stuck-at-faults that must be tested according to the Checkpoint Theorem.
- Show that the two faults d SA0 and g SA1 are equivalent.

Note: Use the following definition for checkpoints: *Checkpoints : Primary inputs and fanout branches of a combinational circuit consisting only of BOOLEAN gates are called the checkpoints.* Thus, expand XOR gate as in BA Figure 4.9, which is same as the one shown below.

This is the new circuit after replacing XOR with elementary gates from BA Figure 4.9:



- AND, OR, NAND, NOR and NOT are called the primitive (same as elementary) Boolean gates. Functions such as XOR are sometimes referred to as complex gates. In the above figure, we have assumed one such implementation. Our result is, therefore, based on this assumption. Other implementations of the XOR function are possible and can give a different set of checkpoints.

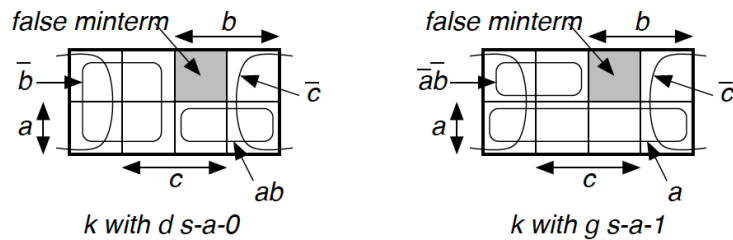
There are nine checkpoints in this circuit. These include three primary inputs, a , b and c , and six fanout branches, $d1$, $d2$, f , $e1$, $e2$ and g . The checkpoint fault set consists of eighteen faults (SA0 and SA1 faults on the nine lines).

Notice that lines d and e of the original circuit are not checkpoints. If we did not model the XOR block with Boolean gates, then those lines will appear to be checkpoints, whose number will be fourteen. However, detection of those faults will not guarantee detection of faults on the fanouts that are internal to the XOR block. Considering the Boolean gate structure, a fault on d corresponds to a simultaneous (multiple) faults on $d1$ and $d2$ and, in general, the detection of a multiple fault is not equivalent to detection of the component faults.

b) We evaluate the output function k corresponding to the two faults:

$$\begin{aligned}
 k(d \text{ s-a-} 0) &= \bar{c} + \bar{b} + \overline{a + b} \\
 &= \bar{c} + \bar{b} + ab \\
 k(g \text{ s-a-} 1) &= \bar{c} + \overline{ab} + \overline{ab} + a \\
 &= \bar{c} + \overline{ab} + a
 \end{aligned}$$

The two faulty functions are shown by Karnaugh maps below. In both cases, the functions have exactly one false minterm, abc . **Since the two faulty functions are identical the corresponding faults are equivalent.**



Note: this type of fault equivalence is functional and is often difficult to find by typical fault analysis tools, which rely on structurally identifiable equivalences.